Experiment no.: 7	Date:

CLOUD COLLABORATION

PRACTICAL SIGNIFICANCE

To understand cloud-based collaboration by using real-time editing features in Google Docs and Google Sheets and to explore version history management.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

PO6: When solving well-defined engineering problems, evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (DK1, DK5, and DK7).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO2: Use Stand-alone and Cloud-based office tools to prepare documents, spreadsheets, and presentations.

THEORY

Cloud Collaboration

Cloud collaboration is the process of allowing multiple users to access, edit, and share files or resources through internet-based platforms.

Advantages:

* *Enables real-time teamwork*

* *Automatic saving of work*

* *Access from any device*

* *Easy sharing of files*

* *Maintains version history*

Examples of cloud collaboration platforms:

** Google Docs*

** Google Sheets*

** Google Drive*

** Microsoft OneDrive*

** Dropbox*

Google Docs

Google Docs is a web-based document editor used to create and edit text documents online.

***Features**:*

** Real-time editing*

** Auto-save*

** Sharing and permissions*

** Comments and suggestions*

** Version history*

***Applications**:*

** Report writing*

** Documentation*

** Group assignments*

** Project work*

Google Sheets

Google Sheets is an online spreadsheet application used to create and analyze data.

***Features**:*

** Spreadsheet calculations*

** Real-time collaboration*

** Formula support*

** Chart creation*

** Version history*

***Applications**:*

** Mark lists*

** Attendance sheets*

** Budget preparation*

** Data analysis*

Version History

Version history stores previous changes made to a document.

***Advantages**:*

** Track changes made by users*

** Restore previous versions*

** Identify contributors*

** Prevent accidental data loss*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Google Docs	Accessed via browser (docs.google.com)	1 per student

6	Google Sheets	Accessed via browser (sheets.google.com)	1 per student
7	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Real-Time Editing and Version History

1. Open a web browser in Ubuntu.

2. Login to Google account.

3. Open Google Docs.

4. Create a new document.

5. Enter sample content.

6. Share the document with another user.

7. Perform collaborative editing.

8. Add comments and suggestions.

9. Open Google Sheets and create sample data.

10. View version history.

11. Restore an earlier version if required.

RESULTS

Thus real-time collaboration using Google Docs and Google Sheets along with version history features was studied and practiced successfully and uploaded the report in Google classroom.

PRACTICAL RELATED QUESTIONS

1. What is cloud collaboration?

2. What is Google Docs?

3. What is Google Sheets?

4. What is version history?

5. What is real-time editing?

SELF- LEARNING COMPONENTS

1. Imagine a team of four students is editing a single Google Sheet for a lab report. One student accidentally deletes a large data table and then closes the browser. Explain how the team can use the 'Version History' tool to see the names of people who edited the file and describe the steps to restore the table back to its original state.

2. In Google Sheets, if three people are working on a project, how can you tell which specific person deleted a value in Cell B5? Name the menu path you would use to find this information and explain how the 'color-coded' system helps identify the person.

3. Create a 1-page overleaf document that includes a title, a mathematical formula, and a list.(Optional Assignment)

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://support.google.com/docs/answer/2494822	Google	Google Docs Help Centre
2	https://support.google.com/docs/answer/190843	Google	Google Docs Help Centre

Experiment no.: 8	Date:

DATA ANALYSIS

PRACTICAL SIGNIFICANCE

To analyze a dataset using Pivot Tables and Slicers in Google Sheets for summarizing and filtering information.

RELEVANT PROGRAM OUTCOMES (POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices

PO2: Identify and analyze well-defined engineering problems and reach substantiated conclusions using codified methods of analysis specific to their field of activity (DK1 to DK4).

PO4: Conduct investigations of well-defined problems by locating and search relevant codes and catalogues, and by conducting standard tests and measurements (DK8).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO3: Apply spreadsheet functions and Generative AI tools to construct interactive dashboards, automated reports, and professional visual presentations for interpreting complex datasets.

THEORY

Data Analysis

Data analysis is the process of organizing and interpreting data to extract useful information.

Applications:

** Student performance analysis*

** Attendance analysis*

** Sales analysis*

** Survey reports*

** Budget planning*

Pivot Table

A Pivot Table is a spreadsheet feature used to summarize and rearrange large amounts of data dynamically.

Functions:

** Calculate totals*

** Calculate averages*

** Count records*

** Group data*

** Rearrange information*

Advantages:

** Faster analysis*

** Easy report generation*

** Better understanding of data*

Slicer

A Slicer is a visual filtering tool used with Pivot Tables.

Functions:

** Filters data interactively*

** Displays selected categories*

** Makes data analysis easier*

Advantages:

** User-friendly*

** Fast filtering*

** Improves visualization*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Google Sheets	Accessed via browser (sheets.google.com)	1 per student
6	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Data Analysis Using Pivot Tables and Slicers

1. Open Google Sheets.

2. Create a blank spreadsheet.

3. Enter sample data.

4. Select the dataset.

5. Create a Pivot Table.

6. Configure row, column, and value fields.

7. Add a Slicer.

8. Filter the data.

9. Analyze the results.

10. Save the spreadsheet.

RESULTS

The dataset was successfully analyzed using Pivot Tables and Slicers in Google Sheets to summarize, organize, and filter information. The observations and analysis results were

documented in a report containing the dataset, Pivot Table output, Slicer-based filtering results, interpretations, and necessary screenshots. The completed report was successfully uploaded and submitted through Google Classroom.

PRACTICAL RELATED QUESTIONS

1. What is a Pivot Table?

2. What is a Slicer?

3. Why is data analysis performed?

4. What is filtering?

5. What is the purpose of Pivot Tables?

6. Name one advantage of Slicers.

7. What is the use of row fields?

8. What is the use of value fields?

9. Which application was used for creating slicers and pivot table?

10. State one application of data analysis.

SELF- LEARNING COMPONENTS

1. Create a Pivot Table from a 50-row sales dataset. Drag "Category" to Rows and "Sales" to Values. Record the Grand Total value displayed.

2. You have a list of sales for 'Laptop', 'Mouse', and 'Keyboard'. Create a Pivot Table to show the total sales for each. Insert a Slicer for 'Region'. If you click 'South' on the Slicer, what happens to the 'Grand Total'? If you add a new row of data to your original list, name the button you must click to update the Pivot Table.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://support.google.com/docs/answer/1272900	Google	Google Sheets Help Centre
2	https://www.geeksforgeeks.org/google-sheets/how-to-create-and-use-pivot-tables-in-google-sheets/	GeeksforGeeks	GeeksforGeeks